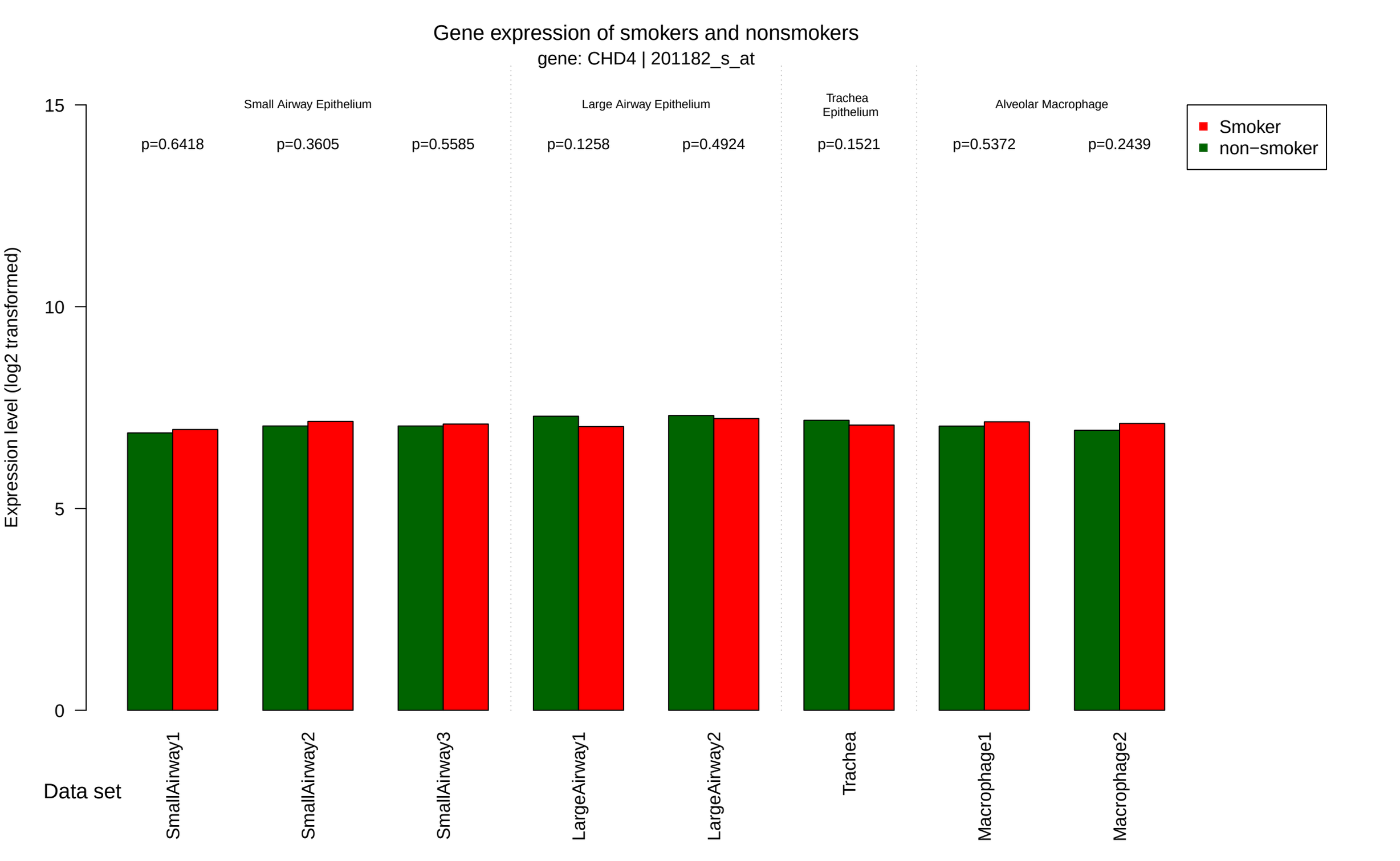
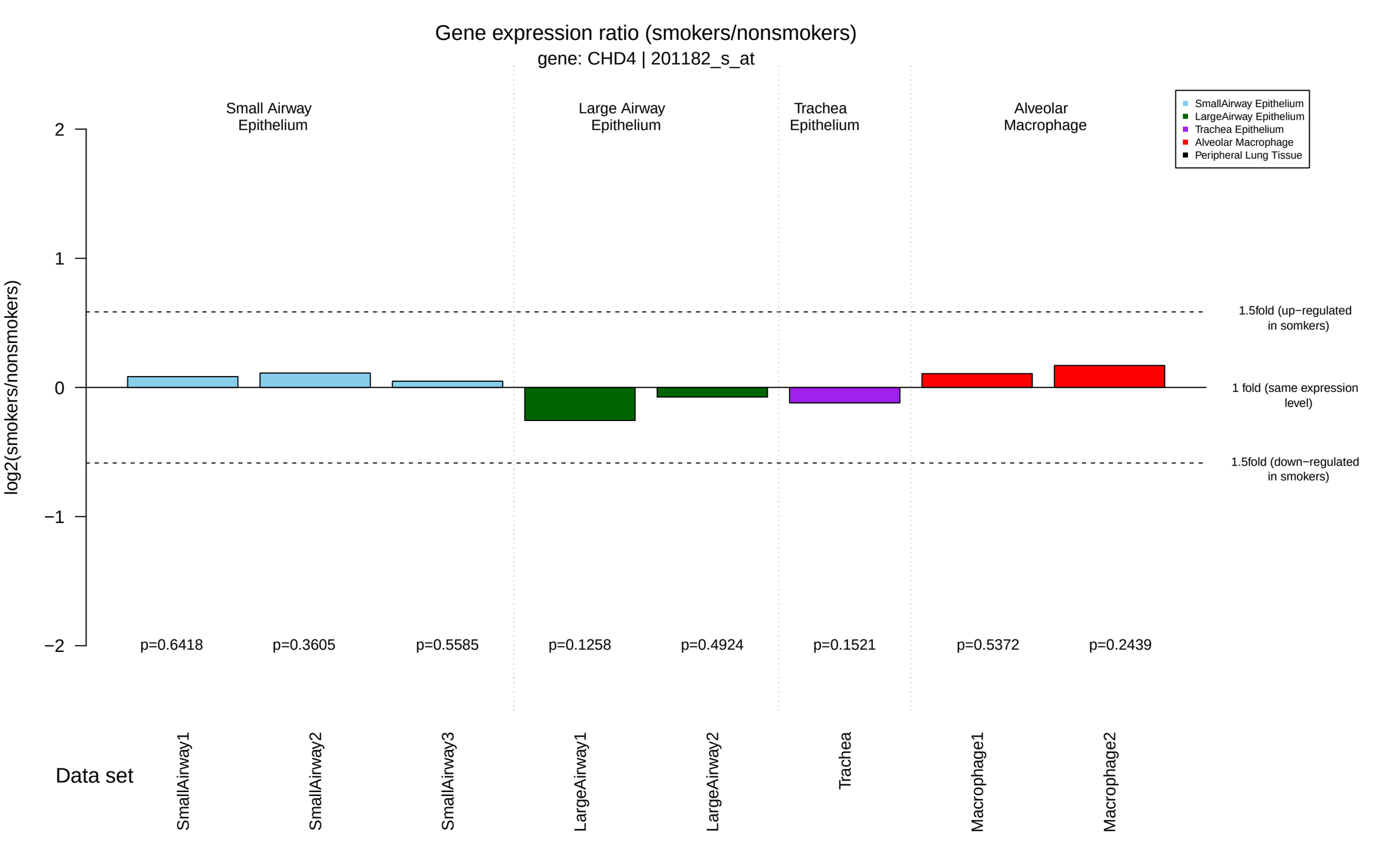
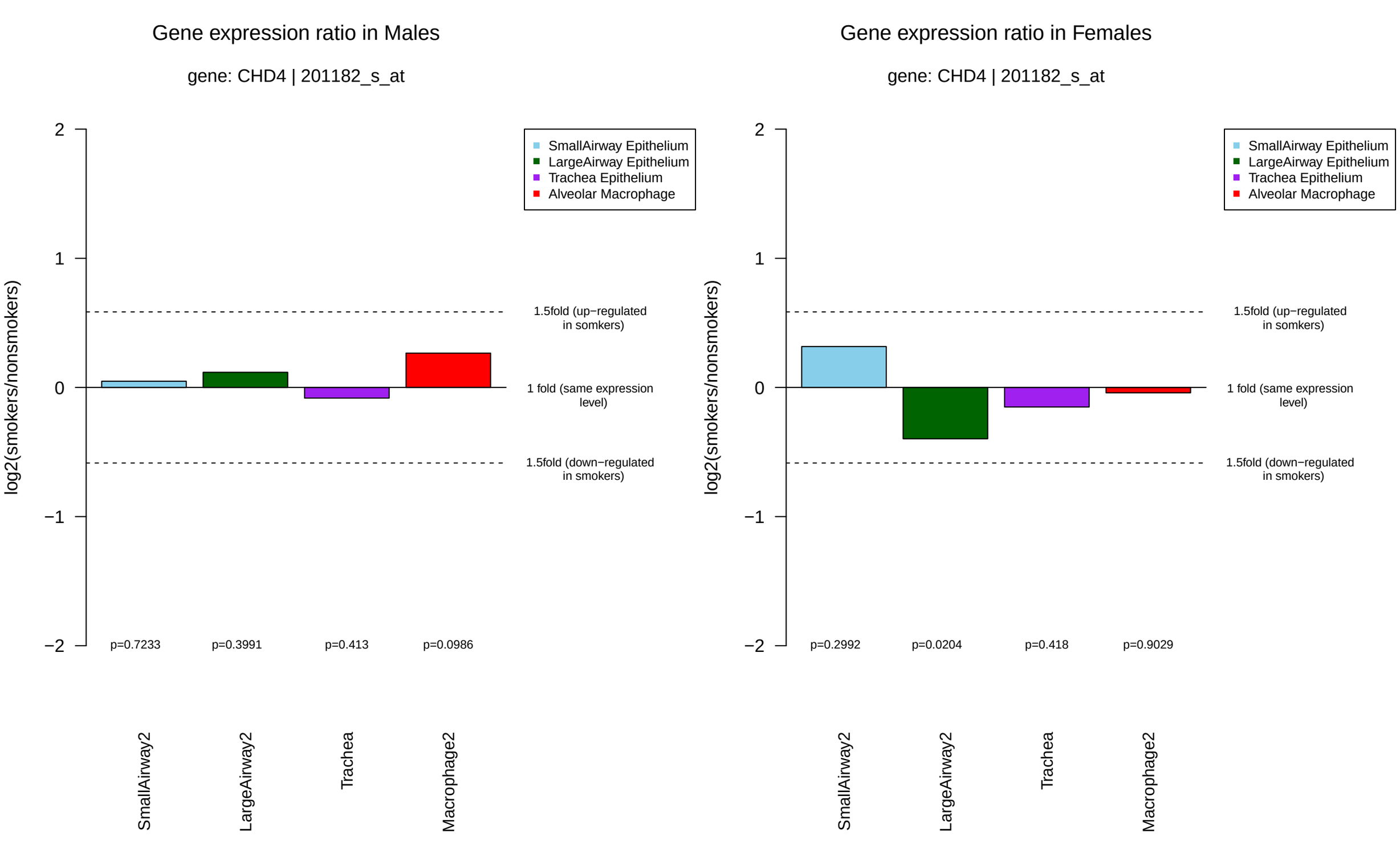
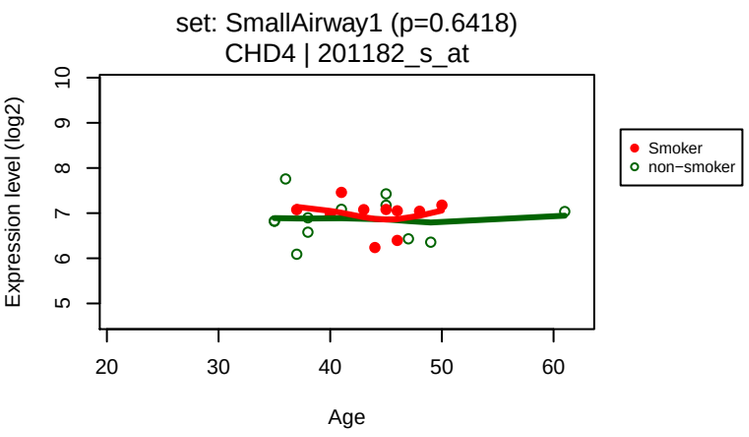
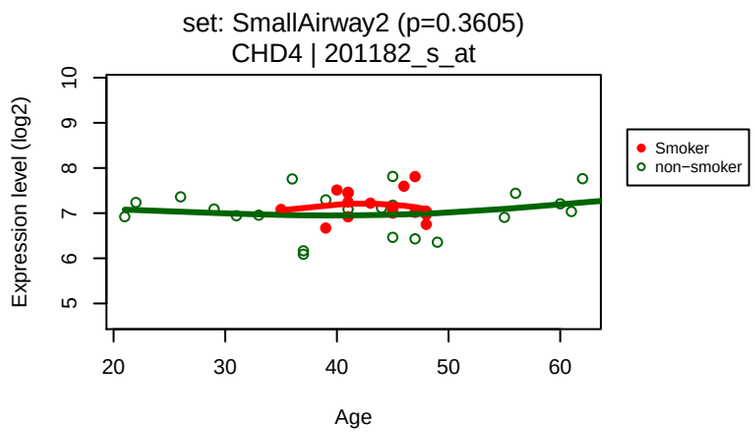
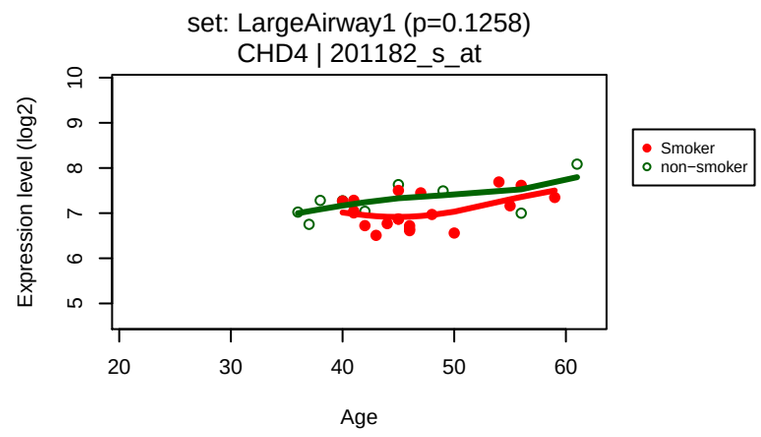

Gene expression of smokers and nonsmokers
gene: CHD4 | 201182_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6418
p=0.3605
p=0.5585
p=0.1258
p=0.4924
p=0.1521
p=0.5372
p=0.2439
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

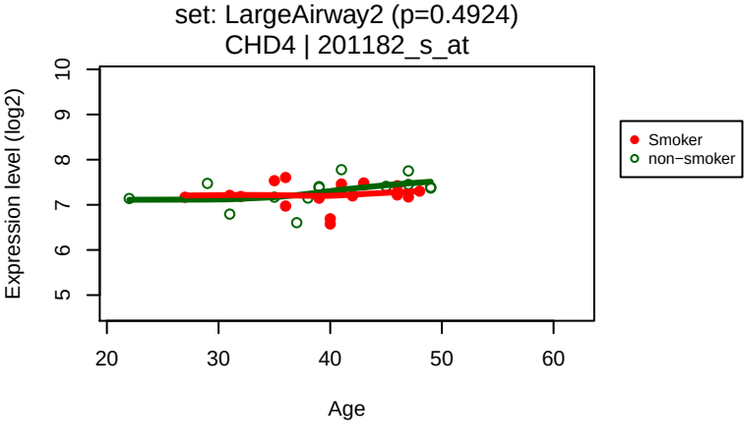

Gene expression ratio (smokers/nonsmokers)
gene: CHD4 | 201182_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6418
p=0.3605
p=0.5585
p=0.1258
p=0.4924
p=0.1521
p=0.5372
p=0.2439
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

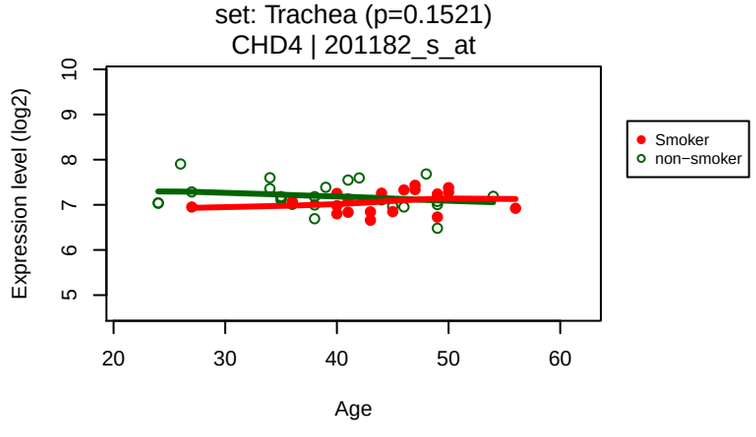

Gene expression ratio in Males
gene: CHD4 | 201182_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7233
p=0.3991
p=0.413
p=0.0986
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHD4 | 201182_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2992
p=0.0204
p=0.418
p=0.9029
SmallAirway2
LargeAirway2
Trachea
Macrophage2

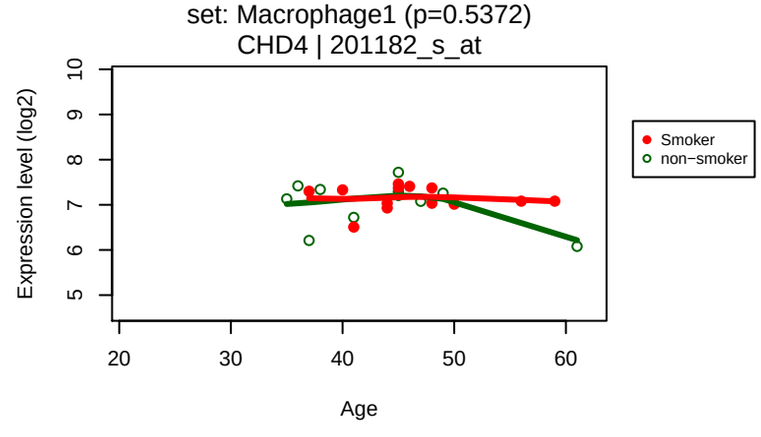

set: SmallAirway1 (p=0.6418)
CHD4 | 201182_s_at
Expression level (log2)
Age
Smoker
non−smoker

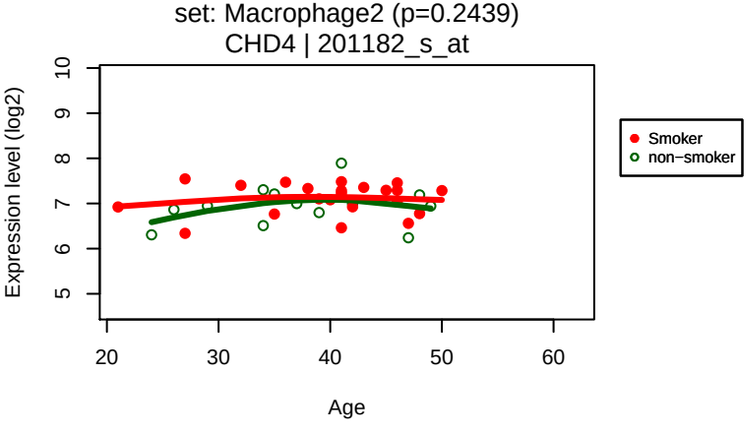

set: SmallAirway2 (p=0.3605)
CHD4 | 201182_s_at
Expression level (log2)
Age
Smoker
non−smoker

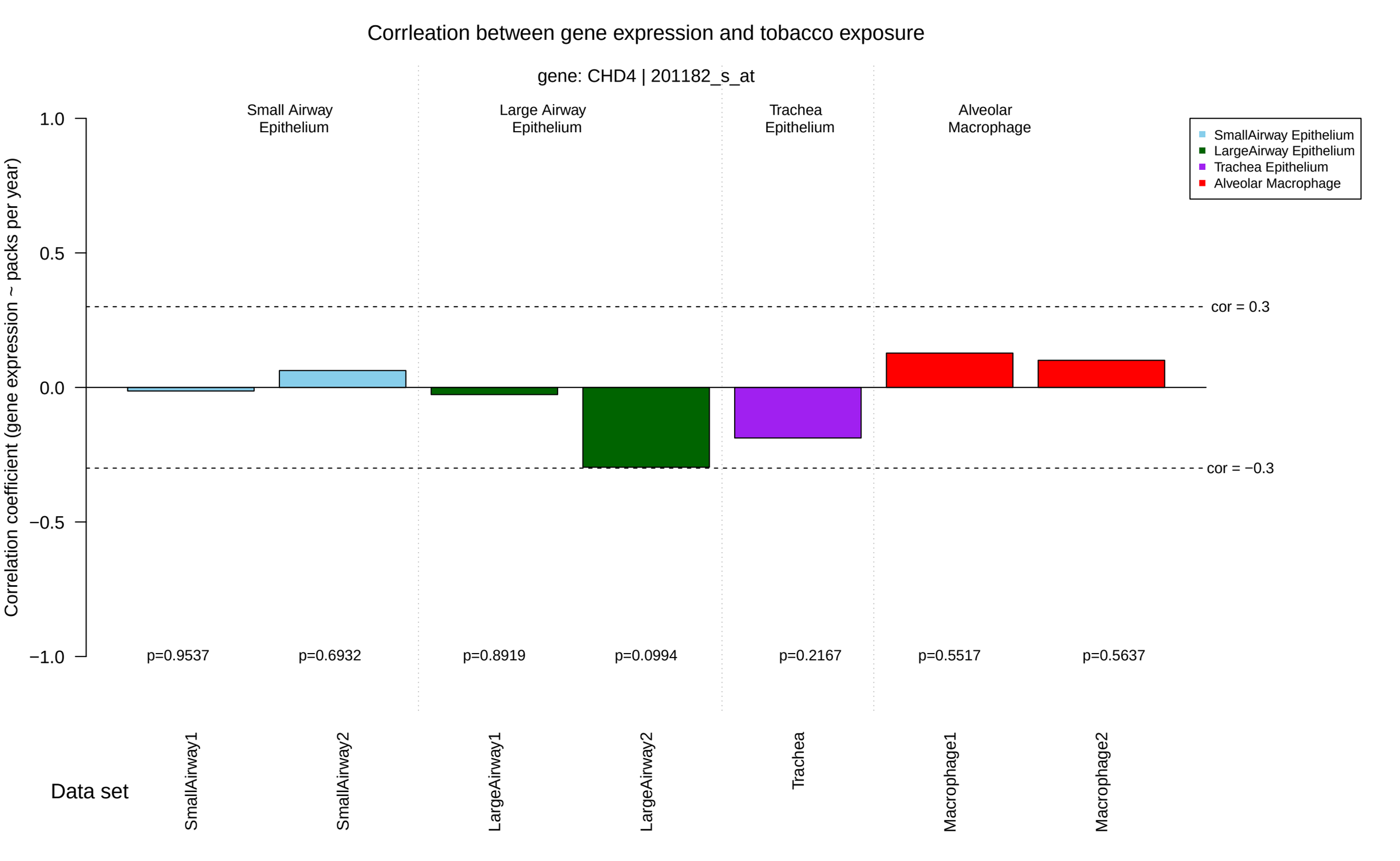

set: LargeAirway1 (p=0.1258)
CHD4 | 201182_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4924)
CHD4 | 201182_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1521)
CHD4 | 201182_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5372)
CHD4 | 201182_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2439)
CHD4 | 201182_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHD4 | 201182_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9537
p=0.6932
p=0.8919
p=0.0994
p=0.2167
p=0.5517
p=0.5637
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2